CAMPUSPILOT AI

A Generative AI-Based Placement Preparation Assistant

PROJECT REPORT

Submitted by: _____
Department: _____
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1. Abstract

CampusPilot AI is a web-based Generative AI application designed to support students during campus placement preparation. The system uses Google's Gemini large language model to analyze resume content, compare candidate skills against a job description, generate technical interview questions, evaluate practice answers, and create personalized study roadmaps. The application provides a simple interface through Streamlit and accepts resume files in PDF or TXT format.

2. Problem Statement

Students often prepare for placements using scattered resources and generic question lists. They may find it difficult to identify resume weaknesses, understand job-specific skill gaps, practice realistic interview questions, and organize preparation time. CampusPilot AI brings these activities together in one accessible assistant.

3. Objectives

• Provide actionable resume feedback without fabricating candidate experience.

• Identify skill overlap and missing skills between a resume and a job description.

• Generate role- and difficulty-specific interview questions with model answers.

• Evaluate a student's draft answer and suggest a clearer response.

• Create a time-bounded, day-wise preparation roadmap.

4. Proposed System and Modules

Module	Description
Resume Analyzer	Extracts PDF/TXT text and asks Gemini to identify strengths, gaps and ATS improvements.
JD Skill Matcher	Compares resume evidence with job requirements and prioritizes missing skills.
Interview Generator	Creates technical questions, concise model answers and follow-up questions.
Answer Evaluator	Scores a practice response, explains missing points and proposes an improved answer.
Study Planner	Builds a day-wise roadmap based on available days, study hours and target role.

5. Technology Stack

Python: core application language. **Streamlit:** interactive web interface. **Google Gemini API:** natural-language generation and reasoning. **PyPDF:** text extraction from PDF resumes. **Requests:** HTTPS communication with the Gemini API. **GitHub:** source-code hosting. **Streamlit Community Cloud:** deployment platform.

6. System Workflow

The user opens the Streamlit application, enters a Gemini API key, uploads a resume where relevant, and selects a module. The app extracts resume text using PyPDF (or reads TXT content), combines it with the user's role, job description, question or planning constraints, and sends a structured prompt to the Gemini API. The API returns generated text, which the application displays in the browser.

7. Implementation Details

The main application is implemented in app.py. The extract_resume function reads PDF or TXT files. The gemini_generate function sends a generateContent request to the Gemini REST endpoint. The run_ai function builds a consistent system prompt that asks the model to be practical, structured and truthful. Streamlit tabs

separate the five user-facing modules.

8. Expected Output

The application produces a resume review, a resume-to-job skill comparison, interview questions with answers, feedback on a candidate's response, and a preparation schedule. Output quality depends on the information supplied and the Gemini model/API availability.

9. Testing

Suggested tests include uploading a text-based PDF resume, uploading a TXT resume, generating questions with only a target role, matching a resume with a sample job description, evaluating a short answer, and generating plans for different preparation durations. Also test missing API keys, invalid keys, network failures and unsupported file types.

10. Limitations and Future Scope

The current version requires an internet connection and a valid Gemini API key. Scanned/image-only PDFs may not yield extractable text. Model responses can contain inaccuracies, so users should verify important technical guidance. Future enhancements may include user accounts, saved preparation history, DOCX support, OCR for scanned resumes, downloadable PDF feedback, speech-based mock interviews, and additional LLM providers.

11. Conclusion

CampusPilot AI demonstrates a practical application of Generative AI in education and career preparation. By combining resume analysis, job matching, interview practice and study planning in one interface, it provides a useful starting point for students preparing for campus recruitment.

12. References

1. Google AI for Developers – Gemini API documentation: https://ai.google.dev/gemini-api/docs

2. Streamlit documentation: https://docs.streamlit.io/

3. PyPDF documentation: https://pypdf.readthedocs.io/